Assignment 2

Week 2

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Problem 1. Normalization by Hand

Normalize vector $|\vec{v_1}\rangle = 2\,|{\uparrow}\rangle + 1\,|{\downarrow}\rangle$ by hand.

Solution. To normalize the vector $|\vec{v_1}\rangle = 2\,|{\uparrow}\rangle + 1\,|{\downarrow}\rangle$, we need to divide it by its norm. The norm of $|\vec{v_1}\rangle$ is given by:

$$||\vec{v_1}\rangle| = \sqrt{|2|^2 + |1|^2} = \sqrt{4+1} = \sqrt{5}$$

Therefore, the normalized vector is:

$$|\vec{v}_{1\,\text{normalized}}\rangle = \frac{2}{\sqrt{5}}\,|{\uparrow}\rangle + \frac{1}{\sqrt{5}}\,|{\downarrow}\rangle \approx \begin{pmatrix} 0.8944 \\ 0.4472 \end{pmatrix}$$

Problem 2. Normalization Using Python

Check using Google Colab. You can use the scikit-learn library to do so:

```python
import numpy as np

v1 = np.array([[2], [1]])
print(v1)

from sklearn.preprocessing import normalize

v2 = v1 / np.linalg.norm(v1)
print(v2)
```

Solution. After running the provided Python code, the output is:

$$|\vec{v}_{1\,\text{normalized}}\rangle \approx \begin{pmatrix} 0.8944 \\ 0.4472 \end{pmatrix}$$

Problem 3. Normalization by Hand with Complex Coefficient

Normalize vector $|\vec{v_1}\rangle = 2\,|{\uparrow}\rangle + (1+i)\,|{\downarrow}\rangle$ by hand.

Solution. The norm of $|\vec{v_1}\rangle = 2\,|{\uparrow}\rangle + (1+i)\,|{\downarrow}\rangle$ is calculated as follows:

$$||\vec{v_1}\rangle| = \sqrt{|2|^2 + |1+i|^2} = \sqrt{4 + (1^2 + 1^2)} = \sqrt{4+2} = \sqrt{6}$$

Thus, the normalized vector is:

$$|\vec{v}_{1\,\text{normalized}}\rangle = \frac{2}{\sqrt{6}}\,|{\uparrow}\rangle + \frac{1+i}{\sqrt{6}}\,|{\downarrow}\rangle \approx \begin{pmatrix} 0.8165 \\ 0.4082 + 0.4082i \end{pmatrix}$$

Problem 4. Normalization Using Python with Complex Coefficient

Verify using Google Colab. Remember you need to use $1 + 1j$ to represent $1 + i$.

Solution. Using the following Python code:

```
v1 = np.array([[2], [1+1j]])
v2 = np.round(v1 / np.linalg.norm(v1), 4)
print(f"Normalized vector:\n{v2}")
```

After running the provided Python code, the output is:

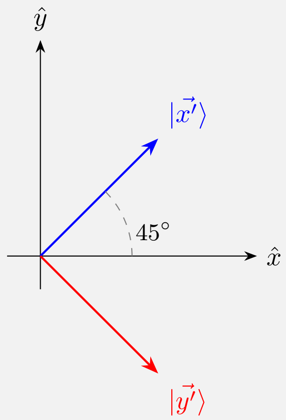
$$|\vec{v_1}_{\text{normalized}}\rangle \approx \begin{pmatrix} 0.8165 \\ 0.4082 + 0.4082i \end{pmatrix}$$

Problem 5. Vectors on a 2D Plane

On a 2D plane, draw the vectors $|\vec{x'}\rangle = \frac{1}{\sqrt{2}}(|\hat{x}\rangle + |\hat{y}\rangle)$ and $|\vec{y'}\rangle = \frac{1}{\sqrt{2}}(|\hat{x}\rangle - |\hat{y}\rangle)$. Write the column forms of $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$. Show that $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$ are orthonormal, i.e. $\langle\vec{x'}|\vec{x'}\rangle = \langle\vec{y'}|\vec{y'}\rangle = 1$ and $\langle\vec{x'}|\vec{y'}\rangle = \langle\vec{y'}|\vec{x'}\rangle = 0$ and . You have derived a new orthonormal basis for a 2D plane. But this is not surprising, right? As $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$ can be obtained by just rotating

Solution. The column forms of $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$, using $|\hat{x}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|\hat{y}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, are

$$|\vec{x'}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad |\vec{y'}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$



Norms:

$$\langle\vec{x'}|\vec{x'}\rangle = \frac{1}{2}(1 \cdot 1 + 1 \cdot 1) = \frac{2}{2} = 1, \qquad \langle\vec{y'}|\vec{y'}\rangle = \frac{1}{2}(1 \cdot 1 + (-1) \cdot (-1)) = \frac{2}{2} = 1$$

Cross terms:

$$\langle\vec{x'}|\vec{y'}\rangle = \frac{1}{2}(1 \cdot 1 + 1 \cdot (-1)) = \frac{1-1}{2} = 0, \qquad \langle\vec{y'}|\vec{x'}\rangle = \frac{1}{2}(1 \cdot 1 + (-1) \cdot 1) = \frac{1-1}{2} = 0$$

Since both vectors have unit norm and their inner product with each other is zero, $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$ form an orthonormal basis.
This is not surprising: $|\vec{x'}\rangle$ and $|\vec{y'}\rangle$ are simply $|\hat{x}\rangle$ and $|\hat{y}\rangle$ rotated counterclockwise by 45°, and a rotation always preserves lengths and angles, so it necessarily maps an orthonormal basis to another orthonormal basis.

Problem 6. Basis Change by Hand

Vector $\vec{v_1} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is in the $|+\rangle\,/\,|-\rangle$ basis. Represent it in the $|0\rangle\,/\,|1\rangle$ basis. Try to convert $|+\rangle$ and $|-\rangle$ in terms of $|0\rangle$ and $|1\rangle$ and perform the substitution. To check your answer, convert it back to $|+\rangle\,/\,|-\rangle$ using the equations introduced in this chapter and you should get back $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

Solution. We have $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Substituting $\vec{v_1} = 1\,|+\rangle + 2\,|-\rangle$:

$$\begin{aligned} \vec{v_1} &= 1\,|+\rangle + 2\,|-\rangle \\ &= 1 \cdot \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) + 2 \cdot \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= \frac{1}{\sqrt{2}}\left[(1+2)\,|0\rangle + (1-2)\,|1\rangle\right] \\ \vec{v_1} &= \frac{3}{\sqrt{2}}\,|0\rangle - \frac{1}{\sqrt{2}}\,|1\rangle \end{aligned}$$

So in the $|0\rangle\,/\,|1\rangle$ basis:

$$\vec{v_1} = \begin{pmatrix} 3/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \approx \begin{pmatrix} 2.1212 \\ -0.7071 \end{pmatrix}$$

Check. Converting back to the $|+\rangle\,/\,|-\rangle$ basis with $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$, and writing $\vec{v_1} = a\,|0\rangle + b\,|1\rangle$ with $a = 3/\sqrt{2}$, $b = -1/\sqrt{2}$:

$$c_+ = \frac{a+b}{\sqrt{2}} = \frac{3/\sqrt{2} - 1/\sqrt{2}}{\sqrt{2}} = \frac{2/\sqrt{2}}{\sqrt{2}} = 1$$

$$c_- = \frac{a-b}{\sqrt{2}} = \frac{3/\sqrt{2} + 1/\sqrt{2}}{\sqrt{2}} = \frac{4/\sqrt{2}}{\sqrt{2}} = 2$$

which recovers $\vec{v_1} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ in the $|+\rangle\,/\,|-\rangle$ basis, exactly as expected.

Problem 7. Vector Normalization by Hand and Using Python

Normalize vector $|\vec{v_1}\rangle = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ (which is in the $|0\rangle\,/\,|1\rangle$ basis) in the $|+\rangle\,/\,|-\rangle$ basis. Use Google Colab to verify your answer.

Solution. From Problem 6, $\vec{v_1}$ expressed in the $|+\rangle\,/\,|-\rangle$ basis (before normalization) is

$$\vec{v_1} = \begin{pmatrix} 3/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

Its norm is

$$|\vec{v_1}| = \sqrt{\left(\frac{3}{\sqrt{2}}\right)^2 + \left(\frac{-1}{\sqrt{2}}\right)^2} = \sqrt{\frac{9}{2} + \frac{1}{2}} = \sqrt{5}$$

so the normalized vector, in the $|+\rangle\,/\,|-\rangle$ basis, is

$$|\vec{v_1}_{\text{normalized}}\rangle = \frac{3/\sqrt{2}}{\sqrt{5}}\,|+\rangle - \frac{1/\sqrt{2}}{\sqrt{5}}\,|-\rangle = \frac{3}{\sqrt{10}}\,|+\rangle - \frac{1}{\sqrt{10}}\,|-\rangle \approx \begin{pmatrix} 0.9487 \\ -0.3162 \end{pmatrix}$$

Using Python to verify:

```
import numpy as np

# v1 = (1, 2) in the |0>/|1> basis
v1_01 = np.array([[1], [2]])

# basis change matrix from |0>/|1> to |+>/|-> coefficients
M = (1 / np.sqrt(2)) * np.array([[1, 1], [1, -1]])

v1_pm = np.round(M @ v1_01, 4)
print("v1 in +/- basis (unnormalized):", v1_pm.ravel())

v1_pm_normalized = np.round(v1_pm / np.linalg.norm(v1_pm), 4)
print("v1 in +/- basis (normalized):", v1_pm_normalized.ravel())
```

which prints $\begin{pmatrix} 0.9487 \\ -0.3162 \end{pmatrix}$, matching the hand-calculated result.

Problem 8. Measurement Probability

For the $\vec{v_1}$ in the last question, what is the probability to find $|+\rangle$ in the $|+\rangle \,/\, |-\rangle$ basis measurement?

Solution. The probability of measuring $|+\rangle$ is the squared magnitude of its coefficient in the normalized vector:

$$P(|+\rangle) = \left| \frac{3}{\sqrt{10}} \right|^2 = \frac{9}{10} = 0.9$$

Problem 9. Vector Normalization by Hand and Using Python 2

Normalize $|\vec{v_1}\rangle$ in the $|0\rangle \,/\, |1\rangle$ basis. What is the probability to find $|0\rangle$ in the $|0\rangle \,/\, |1\rangle$ basis measurement? Use Google Colab to verify your answer.

Solution. The norm of $\vec{v_1} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ in the $|0\rangle \,/\, |1\rangle$ basis is

$$|\vec{v_1}| = \sqrt{1^2 + 2^2} = \sqrt{5}$$

so the normalized vector is

$$|\vec{v_1}_{\text{normalized}}\rangle = \frac{1}{\sqrt{5}} |0\rangle + \frac{2}{\sqrt{5}} |1\rangle \approx \begin{pmatrix} 0.4472 \\ 0.8944 \end{pmatrix}$$

The probability to find $|0\rangle$ is

$$P(|0\rangle) = \left| \frac{1}{\sqrt{5}} \right|^2 = \frac{1}{5} = 0.2$$

Using Python to verify:

```python
import numpy as np

v1 = np.array([[1], [2]])
v1_normalized = v1 / np.linalg.norm(v1)
print("v1 normalized in 0/1 basis:", v1_normalized.ravel())

prob_0 = np.abs(v1_normalized[0, 0]) ** 2
print("P(|0>):", prob_0)
```

which prints $\begin{pmatrix} 0.4472 \\ 0.8944 \end{pmatrix}$ and $P(|0\rangle) = 0.2000$, matching the hand-calculated result.

Problem 10. Basis Change and Measurement

If I get $|1\rangle$ in the measurement in Problem 9, what is the probability of getting $|+\rangle$ if I measure again in the $|+\rangle\,/\,|-\rangle$ basis measurement? (hint: after getting $|1\rangle$, how to represent it in the $|+\rangle\,/\,|-\rangle$ basis?)

Solution. After the measurement collapses the state to $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ in the $|0\rangle\,/\,|1\rangle$ basis, we convert it to the $|+\rangle\,/\,|-\rangle$ basis using $a = 0$, $b = 1$:

$$c_+ = \frac{a+b}{\sqrt{2}} = \frac{0+1}{\sqrt{2}} = \frac{1}{\sqrt{2}}, \qquad c_- = \frac{a-b}{\sqrt{2}} = \frac{0-1}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$$

so $|1\rangle = \frac{1}{\sqrt{2}}\,|+\rangle - \frac{1}{\sqrt{2}}\,|-\rangle$, which is already normalized. The probability of getting $|+\rangle$ is

$$P(|+\rangle) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} = 0.5$$

APPENDIX 1

Assignment 02 | Code Solution

assignment-02-code

August 28, 2026

1 Assignment II

1.1 Import External Libraries

[1]:
```
import numpy as np
import sympy as sp

from IPython.display import display, Math
from sklearn.preprocessing import normalize
```

[2]:
```
BRAKET_01_BASIS = "\\vert 0 \\rangle / \\vert 1 \\rangle"
BRAKET_PM_BASIS = "\\vert + \\rangle / \\vert - \\rangle"
```

1.2 2. Normalization Using Python

[3]:
```
v1 = np.array([[2], [1]])
display(Math(f"\\text{{Original Vector: }} {sp.latex(sp.Matrix(v1))}"))

v2 = np.round(v1 / np.linalg.norm(v1), 4)
display(Math(f"\\text{{Normalized Vector: }} {sp.latex(sp.Matrix(v2))}"))
```

$$\text{Original Vector: } \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\text{Normalized Vector: } \begin{bmatrix} 0.8944 \\ 0.4472 \end{bmatrix}$$

1.3 4. Normalization Using Python with Complex Coefficient

[4]:
```
v1 = np.array([[2], [1+1j]])
v2 = np.round(v1 / np.linalg.norm(v1), 4)
display(Math(f"\\text{{Normalized Vector: }} {sp.latex(sp.Matrix(v2))}"))
```

Normalized Vector: $\begin{bmatrix} 0.8165 \\ 0.4082 + 0.4082i \end{bmatrix}$

1.4 7. Vector Normalization by Hand and Using Python

[5]:
```
# v1 = (1, 2) in the |0>/|1> basis
v1_01 = np.array([[1], [2]])

# basis change matrix from |0>/|1> to |+>/|-> coefficients
M = (1 / np.sqrt(2)) * np.array([[1, 1], [1, -1]])

v1_pm = np.round(M @ v1_01, 4)
display(Math(f"v_1 \\text{{ in }} {BRAKET_PM_BASIS} \\text{{ basis␣
↪(unnormalized):}}"))
display(Math(sp.latex(sp.Matrix(v1_pm.ravel()))))

print("---------------")

v1_pm_normalized = np.round(v1_pm / np.linalg.norm(v1_pm), 4)
display(Math(f"v_1 \\text{{ in }} {BRAKET_PM_BASIS} \\text{{ basis (normalized):
↪}}"))
display(Math(sp.latex(sp.Matrix(v1_pm_normalized.ravel()))))
```

v_1 in $|+\rangle/|-\rangle$ basis (unnormalized):

$$\begin{bmatrix} 2.1213 \\ -0.7071 \end{bmatrix}$$

```
---------------
```

v_1 in $|+\rangle/|-\rangle$ basis (normalized):

$$\begin{bmatrix} 0.9487 \\ -0.3162 \end{bmatrix}$$

1.5 9. Vector Normalization by Hand and Using Python 2

[6]:
```
v1 = np.array([[1], [2]])
v1_normalized = np.round(v1 / np.linalg.norm(v1), 4)
display(Math(f"v_1 \\text{{ in }} {BRAKET_01_BASIS} \\text{{ basis (normalized):
↪}}"))
display(Math(sp.latex(sp.Matrix(v1_normalized.ravel()))))

print("---------------")

prob_0 = np.abs(v1_normalized[0, 0]) ** 2
display(Math(f"P(\\vert 0 \\rangle) = {prob_0:.4f}"))
```

3

v_1 in $|0\rangle/|1\rangle$ basis (normalized):

$$\begin{bmatrix}0.4472\\0.8944\end{bmatrix}$$

$P(|0\rangle) = 0.2000$